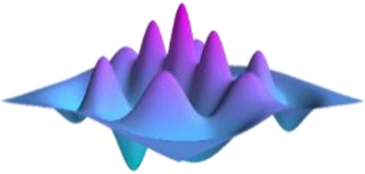
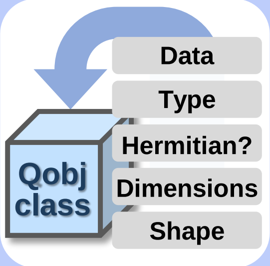
Overview of QuTiP



1 QuTiP - Core

Qobj



Data Layers
Dense, CSR (compressed sparse row) and DIA (diagonal)

Functions
Eigenstates, Matrix elements, Norms, etc.

Utility Functions
Entanglement measures, Distances,
Superoperator representations (Choi, Kraus, etc.), Channels, ENR states,
Two-time correlation functions and spectra, MPI support, etc.

Environment class
Flexible environment parameterization (Bosonic and Fermionic)

Solvers

mesolve
Lindblad master equation

mcsolve + nm_mcsolve
Monte-Carlo master equation

brmesolve
Bloch-Redfield master equation

fmmesolve
Floquet master equation

krylovsolve
Krylov subspace solver

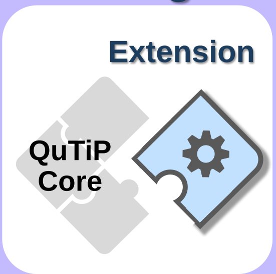
smesolve
Stochastic master equation

HEOMsolver
Hierarchical equations of motion

PIQS
Permutational invariant systems

2 QuTiP - Packages

Packages



QIP: Pulse-based Quantum Circuit Simulator
allows circuits to be run on different hardware backend simulations at the level of time-dependent pulses and noise.

QOC: Quantum Optimal Control Package
supports for CRAB, GRAPE and GOAT algorithms.

JAX: JAX Data Layer
supports the popular JAX package, allowing for GPU and autograd.